

IRIS DATA VISUALIZATION

Synent Technologies Data Science Internship

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Objective

The objective of this project is to visualize patterns, distributions, and relationships in the Iris dataset using various graphical techniques. The visualizations help in understanding species characteristics and feature behavior within the dataset.

Dataset Information

Dataset Name: Iris Dataset

Source: UCI Machine Learning Repository / Scikit-Learn

Rows: 150

Columns: 6

Features:

- Id
- SepalLengthCm
- SepalWidthCm
- PetalLengthCm
- PetalWidthCm
- Species
- Methodology

The following visualization steps were performed:

- Loaded the Iris dataset using Pandas
- Examined dataset structure and basic information
- Checked for missing values
- Created a Bar Chart to visualize species distribution
- Generated a Histogram to analyze Sepal Length distribution
- Created a Scatter Plot to study the relationship between Sepal Length and Petal Length
- Compared average feature values across species using a Feature Comparison Bar Chart
- Results

The visualizations were successfully generated.

Key observations include:

- All three species contain an equal number of samples.
- Sepal Length values follow a clear distribution pattern.
- Setosa species is distinctly separated from the other species in the scatter plot.
- Versicolor and Virginica show some overlap in feature values.
- Petal measurements provide strong differentiation among species.

Generated Visualizations:

- Bar Chart
- Histogram
- Scatter Plot
- Feature Comparison Chart
- Conclusion

Data visualization is an important step in exploratory data analysis. The Iris dataset was successfully visualized using multiple chart types, providing meaningful insights into species distribution, feature relationships, and differences among flower species. These visualizations make the dataset easier to understand and prepare for future machine learning tasks.